UNIVERSITY OF ALBERTA

Engineering
Co-op Program

Kalen Pare

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TECHNICAL SKILLS

Mechanical: SolidWorks | Fusion | CAD | DFM | GD&T | DfAM | 3D Printing | Rapid Prototyping
Software & Embedded: Python | C++ | Git | Circuit Design | Soldering | Raspberry Pi | Arduino

WORK EXPERIENCE

Production Lead - RoBIM Technologies Inc. May 2026 - Dec. 2026

- Led daily operations of RoBIM's Robotic Factory-as-a-Service (RFaaS) fabrication hub, overseeing production planning, resource allocation, and execution.
- Bridged product specifications, mechanical design, process optimization, and shop-floor operations to translate engineering requirements into efficient manufacturing workflows.
- Implemented an enterprise resource planning (ERP) system to track production KPIs, improve operational visibility, and support data-driven decision-making.
- Managed production scheduling and supervised a team of 5+ labourers and co-op students, supporting seven-figure production revenue.

Mechatronics Intern - RoBIM Technologies Inc. Jan. 2026 -Dec. 2026

- Designed, fabricated, and prototyped custom robotic end-of-arm tooling (EOAT), integrating CNC-machined, waterjet-cut, and 3D-printed components with pneumatic actuators and mechanical linkages.
- Followed iterative design cycles, focusing on critical system modules to deliver a minimum viable product (MVP)
- Designed, assembled, and wired an industrial control cabinet incorporating PLCs, I/O modules, and relays.
- Programmed ABB IRB 6700 robots to execute complex fabrication tasks in a live production environment.

Robotic Fabrication Intern - RoBIM Technologies Inc. Apr. 2025 - Aug. 2025

- Designed a robotic end-effector for adhesive dispensing, integrating sensor feedback with pneumatic actuation.
- Achieved a 54.5% reduction in adhesive cycle time by fine-tuning dispensing parameters and performing failure mode analysis, ensuring superior bead quality through systematic validation.
- Designed and built a universal framing jig that improved frame consistency while reducing cycle time by 60%.
- Conducted time studies throughout the process to identify critical bottlenecks, driving data-backed decisions.

PROJECT EXPERIENCE

Robotic Cycloidal Actuator - (Passion Project) Oct. 2025 - Present

- Designed a complex 100-part electromechanical assembly in SolidWorks, integrating components from my motor.
- Achieved a 15:1 reduction with a 55mm axial length, maximizing torque density in a compact form factor.
- Currently working on electrical integration; upcoming testing will characterize torque and efficiency.

Outrunner Brushless DC Motor (BLDC) - (Passion Project) Aug. 2025 - Oct. 2025

- Prototyped the components using iterative 3D printing to achieve tight tolerances for bearings and magnets.
- Integrated a laminated stator core and hand-wound coils in a Wye configuration to optimize flux distribution.
- Conducted back-EMF testing (Kv = 443.4 RPM/V), identifying the need for a machined rotor or steel flux ring.

EDUCATION

BSc Mechatronics Engineering (Co-op) - University of Alberta 2024 - Present

- Availability: 4-12-month co-op term beginning Fall 2027 | GPA: 3.8/4.0